MILITARY SPECIFICATION

MOUNT, HOWITZER AND GUN, M158 AND M174

This Specification is approved for use within the U. S. Army Armament, Munitions and Chemical Command, and is available for use within all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of hydropneumatic gun mount for use in combat vehicles.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following Specification and Standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

BB-N-411	-Nitrogen Type 1, Class 1, Grade B or Type 1, Class 2, Grade A

MILITARY

MIL-H-6083	-Hydrualic Fluid, Petroleum Base for Preservation and Testing
MIL-G-10924	-Grease, Automotive and Artillery
MIL-w-63150	-Weapons and Support Materiel, Standard Quality Assurance Provisions For

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament, Munitions and Chemical Command, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

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STANDARDS

MILITARY

MIL-STD-109	-Quality Assurance Terms and Definitions
MIL-sTD-130	-Identification Marking of U.S. Military Property

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

D10891986	-Cylinder Assembly, Recuperator & Counter Recoil
K10891937	-Mount, Howitzer and Gun (M158)
K10892030	-Mount, Gun Installation (M158)
K12253871	-Installation, Gun Mount
K12253872	-Mount, Gun; Howitzer (M174)

PUBLICATIONS

P10891937	-Special Packaging Instructions for Mount, Howitzer and Gun
P12253872	-Special Packaging Instructions for Mount, Howitzer and Gun

(COpies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of Precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4 and 6.2).

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3.2 Materials. Materials shall confrom to the material specifications referenced on the applicable drawings. Certification of material may be accepted in lieu of material tests. The contractor shall be responsible for determining that materials, assemblies, and parts requiring product qualifications are listed or qualified for inclusion on the applicable Qualified Products List (QPL). In the instance that a qualified vend or no longer exists on the QPL, the contractor is responsible for selecting a vendor whose product meets the minimum requirement of the specification. The contractor must also submit to the government for approval documentation which confirms the vendors ability to meet the specification in question. Contractors inspection record shall specifically list such components, name of Supplier{s) , and number and date of applicable QPL or government approved vendors not on a QPL.

3.3 Construction. The mount, and all of its assemblies and components shall be constructed and assembled as specified on the applicable drawings referenced on Drawings K10892030, K10891937, K12253871 and K12253872. Dimensions and tolerances shall be as specified on the drawings.

3.3.1 Hydraulic system. The recoil mechanism shall be filled with hydraulic fluid, bled and refilled until all air and measurable mositure are eliminated. The initial and subsequent filling of the hydraulic system shall be with hydraulic 0il conforming to MIL-H-6083 and filtered through a 5 micron filter Within five minutes after mechanical gymnastication, a minimum of one pint of oil shall be removed from the lowest drain hole from each recoil cylinder. Sample shall be drained into a suitable container. Particles in each sample shall not exceed the limits specified in Table I.

TABLE I. Particle limits.

Particles in either sample shall not exceed the following:

Metallic

Ferrous - 40 microns and a maximum of two particles over 40 microns but not exceeding 200 microns in greatest dimension.

Non-Ferrous - 200 microns and a maximum of two particles over 200 microns but not exceeding 400 microns in greatest dimension.

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Non-Metallic

Abrasive - 200 microns with maximum of two particles over 200 microns but not exceeding 400 microns in greatest dimension.

Non-abrasive - No limits, however, close surveillance to be maintained to insure minimal quantity.

Lint/Fibers - 600 microns X 60 microns or a maximum of eight particles over 600 microns x 60 microns but not exceeding 3000 microns x 200 microns.

3.3.2 Surface cleanliness. All finish machined surfaces shall be free of condensation or other foreign residues from the time of final machining until final assembly. All parts and assemblies being assembled into the completed mount shall be free of all foreign material after assembly. Assembly shall be performed in the controlled environment specified in 6.3.

3.3.3 Part modification. No parts shall be altered or handfitted during assembly unless such modification has been authorized by an approved drawing change, or is authorized by instructions presently listed on the drawings.

3.3.4 Air, gas. and accumulator system. The gas system shall be clean and free o-f all foreign matter. The hydroscopic content of the air used to blow the lines shall not exceed one percent of the total volume of the air used. The system shall be filled, bled, and refilled until all air and measurable moisture are eliminated.

3.3.5 Cylinder pressure. The cylinders of the mount shall withstand the pressures specified on the drawings without damage, permanent deformation or leakage.

3.3.6 Pistons, rods, and nuts or collars. The pistons, piston rods, and nuts or collar shall withstand the pull test specified on the drawings without permanent set or failure of the screw threads.

3.3.7 Lubrication. Lubricants conforming to MIL-G-10924 shall be used for the assembled mount.

3.3.8 Gas. The nitrogenn used to charge and fill the nitrogen chamber shall conform to BB-N-411. The charging pressure shall be specified on Drawing 10891986.

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3.4 Performance.

3.4.1 Mechanical gymastication. Each mount shall be subjected to the mechanical gymnastication test as specified in 4.6.2 without evidence of erratic movement and shall meet the requirements of 3.4.1.1 and 3.4.1.2. After gymnastication, the gun mount shall meet the requirement of 3.4.5.

3.4.1.1 Counter recoil time. The maximum counter-recoil time shall be 3.5 seconds at 0° elevation and 4 seconds at 650 elevation after breakin.

3.4.1.2 Operating rods. All operating rods shall be inspected for scratches during and after mechanical gymnastication is completed. Continued evidence of rod scratching will result in teardown of the affected cylinder, polishing out the scratch or replacing of the rod, replacing of the seal and cleaning and inspection of the cylinder assembly components. Light scratches that can be polished out and that do not re-occur with continued cycling are not cause for teardown.

3.4.2 Functional firing. The Gun Mount shall withstand the functional firing test as specified in 4.6.3 without evidence of malfunction or failure and shall meet the criteria of 3.4.2.1 through 3.4.2.3. After functional firing, the gun mount shall meet the requirement of 3.4.5, 3.4.6, 3.4.7 and 3.4.8.

3.4.2.1 Recoil travel. The recoil travel (in inches) shall be as follows:

Elevation in Degrees (Simulated or Actual)	Travel	
	Minimum	Max imum
0° to 8°	62.8	70.5
27° to Maximum	27.5	32.9

3.4.2.2 Total cycle time. Total cycle time shall not be less than 1 second and not more than 4 seconds.

3.4.2.3 Recoil performance zones. For all 100% acceptance cycles, actual recoil rod pull vs. time curves shall be within defined recoil performance zones shown in Figures 1 and 2.

3.4.3 Proof firing. Each sample mount shall be subjected to proof firing at a Government proving ground as specified in 4.6.4 without evidence of malfunction or failure and shall meet the criteria of 3.4.3.1 and 3.4.3.2. After proof firing, the gun mount shall meet the requirements of 3.4.5, 3.4.6, 3.4.7 and 3.4.8.

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3.4.3.1 Recoil travel. The recoil travel (in inches) shall be as follows:

Elevation in Degrees	Travel	
	Minimum	Maximum
0° to 80	62.8	71
27° to Maximum	27.5	32

3.4.3.2 Total cycle time. Total cycle time shall not be less than 1 second and not more than 4 seconds.

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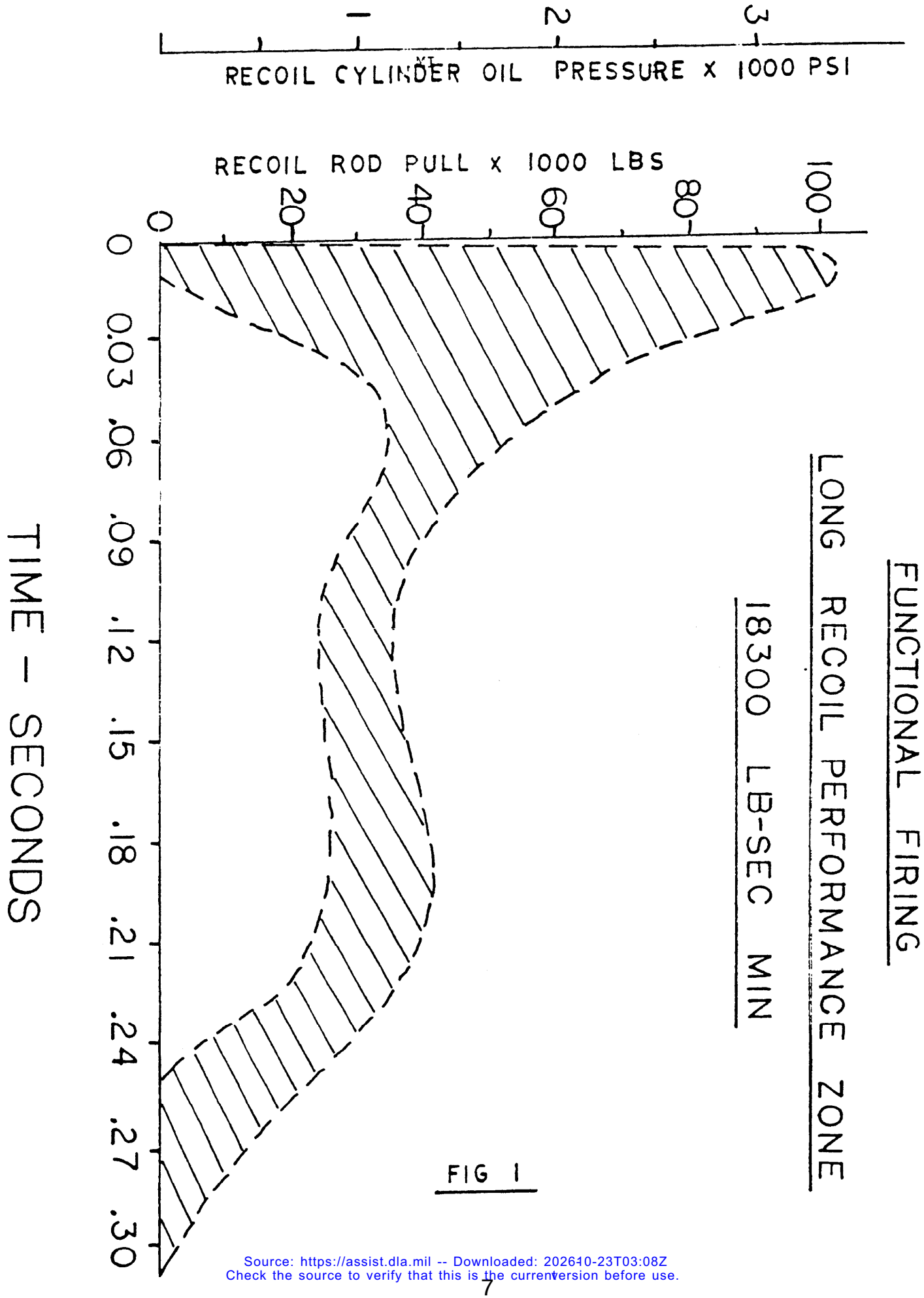


FIG 1

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FUNCTIONAL FIRING

SHORT RECOIL PERFORMANCE ZONE

WITHOUT ELEVATION SIMULATION

18300 LB·SEC MIN

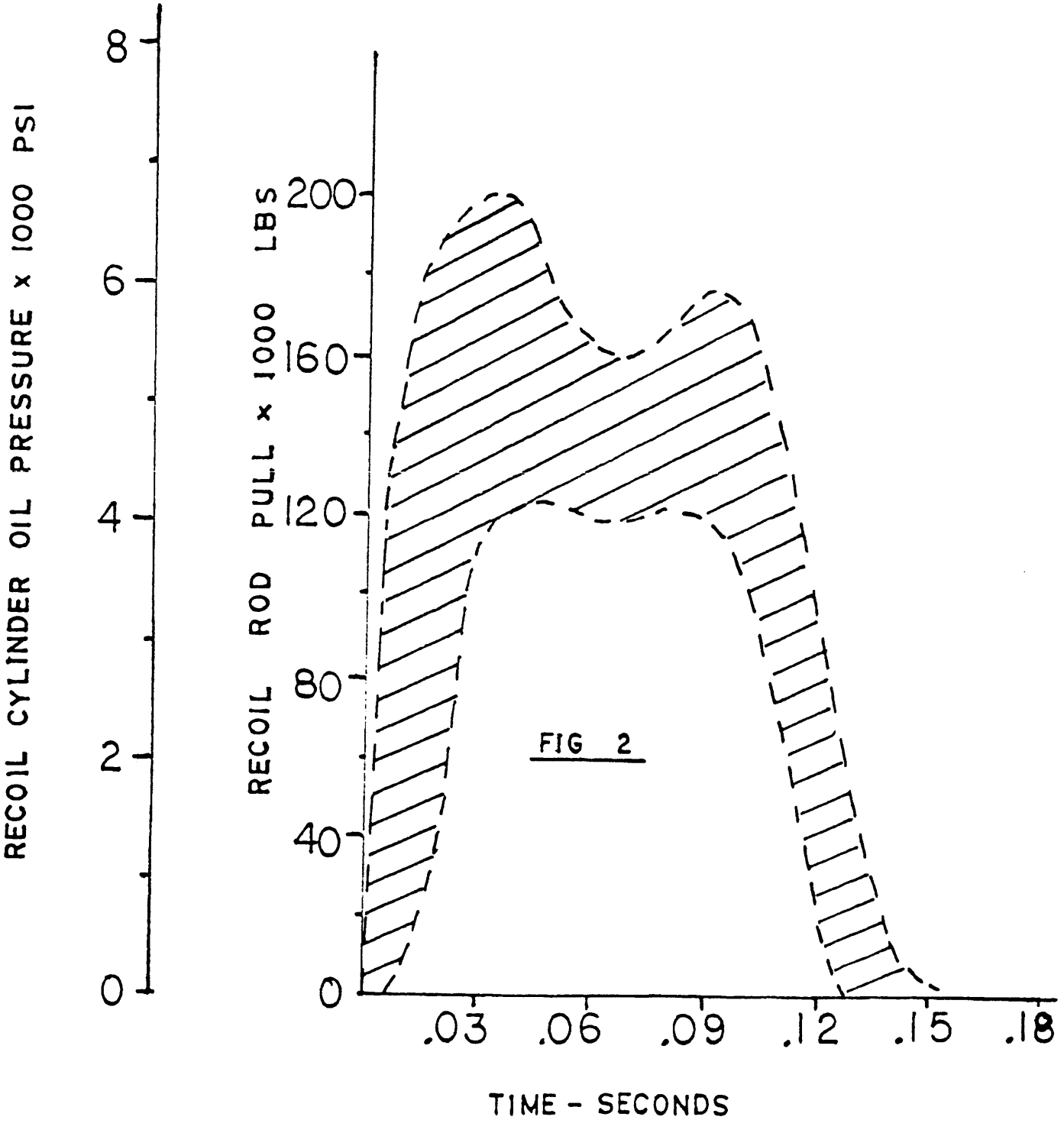


FIG 2

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3.4.3.3 Recoil performance zones. For all full charge firings, actual recoil rod pull vs. time curves shall be within defined recoil performance zones shown in Figures 3 and 4.

3.4.4 Climatic conditions. A sample as specified in 4.5.1 shall be proof fired as specified in 3.4.3 at each of the following temperatures: +125 F and -50°F.

3.4.5 Hydraulic leakage. During and immediately after mechanical gymnastlcatlon, functional firing, and proof firing. there shall be no visible oil leakage from the recoil mechanism.

3.4.6 Nitrogen leakage. Mounts shall also be examined for leakage of nitrogen. Any - indications of foamy or emulsified condition of reserve oil indicates gas leakage and shall be cause for rejection.

3.4.7 Correlation of data. The data recorded for 3.4.2, functional firing, and for 3.4.3, proof firing, shall be correlated. If the correlation etween length of recoil and cycle time is not satisfactory, as determined by the procuring agency, the mounts shall be returned to the contractor.

3.4.8 Packings. Packings shall be adjusted so that the mount shall function with a minimum of friction. The friction of the floating piston and recoil rod and counter recoil piston rod shall be within the following limits after assembly, following mechanical gymnastication, functional firing and proof firing.

Part Name	Pressure Pounds Per Square Inch (psi)	
Floating Rod and counter recoil piston rod	30 Min. 50 Min.	250 Max. 250 Max.

The buffer must be five (5) inches out of battery when the friction values are taken.

3.5 Protective finishes.

3.5.1 Inorganic. Inorganic finishes, including metal deposited coatings and surface treatments, shall be as specified on the drawings.

3.5.2 Painting. Painting shall be as specified on the drawings.

3.6 Marking. Identification marking, including application of part numbers, and instruction marking shall be as specified on the drawings and in MIL-STD-130:

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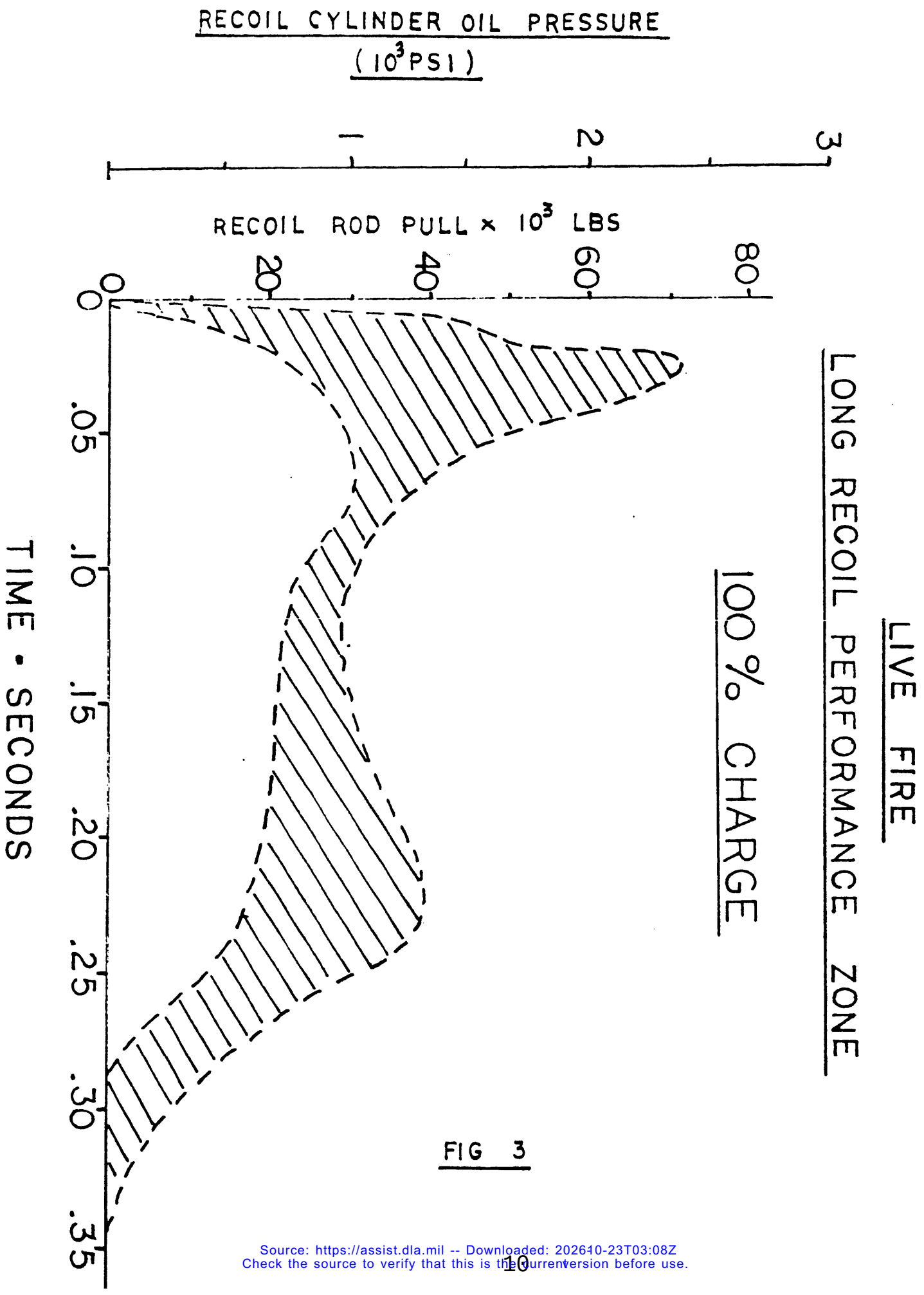


FIG 3

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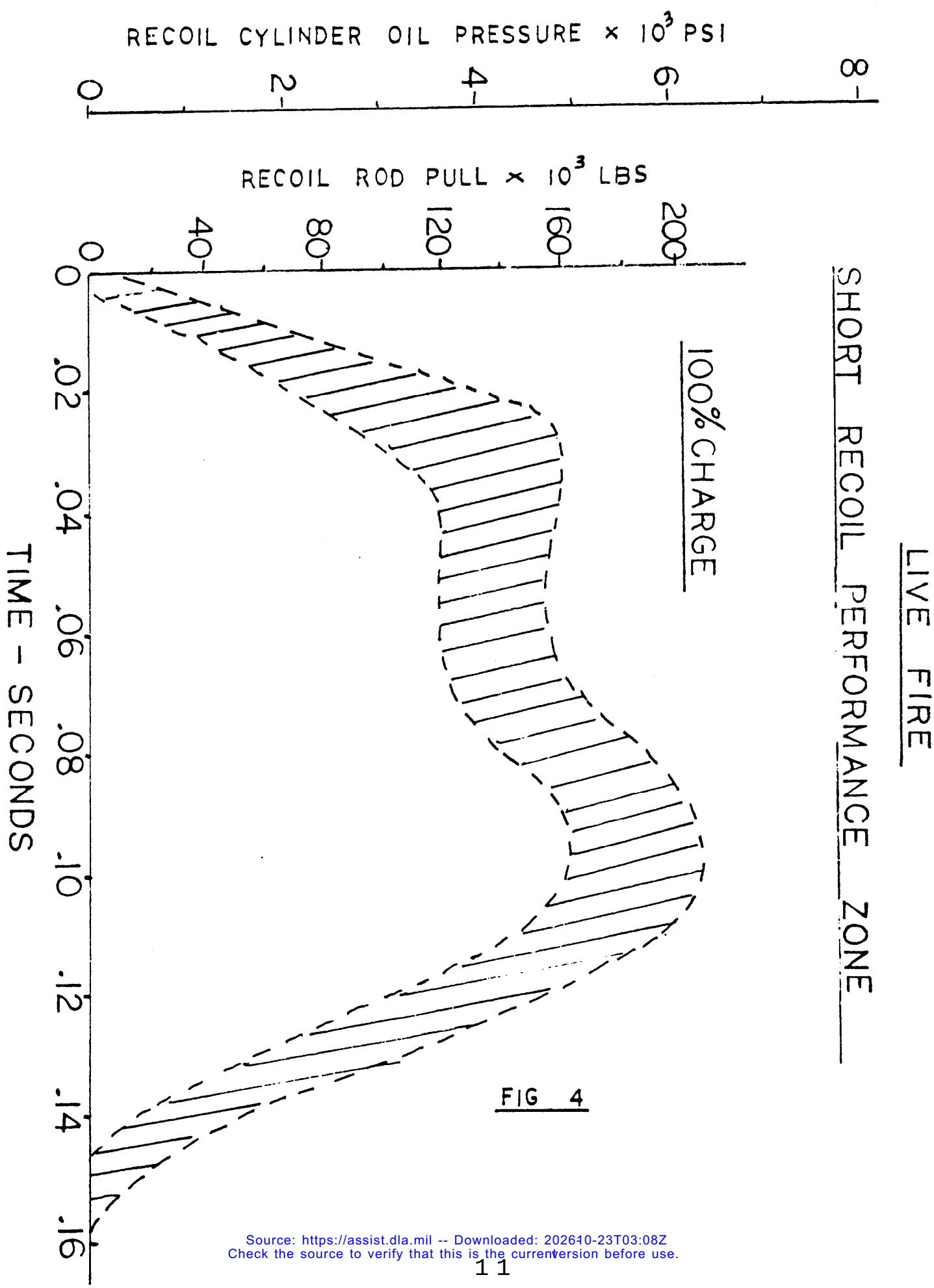


FIG 4

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3.7 Data requirements. The contractor shall generate data in accordance with the requirements of the data item description cited in 4.6.5.

3.8 Workmanship. Each mount shall be free of defects that affect overall appearance, serviceability, or performance such as burrs, fins, sharp edges, and surface flaws or blemishes.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet the requirements of Section 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality proqram. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Quality assurance terms and definitions. Quality assurance terms and definitions as used in this specification are in accordance with MIL-STD-109.

4.3 Classification of inspection. The inspection requirements specified are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

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4.4 First aritcle.

4.4.1 First article sample. A first article sample shall be selected from early production and submitted for testing at a location approved by-the Government in accordance with the contract requirements (see 6.2). The sample shall be representative of the production processes to be used during quantity production.

4.4.2 First article inspection. The first article sample inspection shall be inspected to determine conformance with the requirements of this specification.

4.4.3 First article rejection. If any sample fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of the sample to comply with the requirements of this specification.

4.5 Quality conformance inspection.

4.5.1 Sampling. Each mount shall be inspected for compliance with the specification except sampling for proof firing as specified in 4.7.4 shall be the first three production mounts, and one from each two month's production or one from each twenty-five produced. In addition, the first three production mounts shall be climatically fired as specified in 3.4.4. Failure to comply with the specified requirements shall be cause for rejection.

4.5.2 Examination.

4.5.2.1 Materials. Materials used in the manufacture of mounts shall be examined and tested in accordance with the material specifications referenced on the applicable drawings. The contractor's inspection records shall be checked to determine conformance to 3.2.

4.5.2.2 Hydraulic system. To determine compliance with 3.3.1, a minimum of one pint of oil shall be removed from each recoil cylinder of each mount. The sample shall be taken within five minutes after gymnastication. Evidence of particles exceeding 3.3.1 shall be cause for rejection.

4.5.2.3 Surface cleanliness. Each mount shall be examined to determine compliance with the cleanliness requirements specified in 3.3.2. Failure to comply with 3.3.2 shall be cause for rejection of the mount.

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4.5.2.4 Part modification. Each mount shall be examined to determine compliance with requirements specified in 3.3.3. Any evidence of parts being altered or hand-fitted during assembly beyond the provisions of 3.3.3, shall be cause for rejection of the mount.

4.5.2.5 Gas and accumulation system. To determine conformance to 3.3.4, the system shall be examind for cleanliness, air and moisture. Failure to comply with 3.3.4 shall be cause for rejection of the mount.

4.5.2.6 Cylinder pressure. To determine compliance with 3.3.5, the cylinders of each mount shall be pressurized in accordance with drawing requirements and inspected for damage, permanent deformation or leakage. Pressures shall be meassured with Government-approved gages. Failure to comply with 3.3.5 shall be cause for rejection of the mount.

4.5.2.7 Pistons, rods and nuts or collars. To determine compliance with 3.3.6, the pistons, piston rods, and nuts or collars of each mount shall-be inspected for permanent set or damaged threads after being pull-stressed by Government approved equipment. Failure to comply with 3.3.6 shall be cause for rejection of the mount.

4.5.2.8 Lubrication. Inspection and tests for lubricants shall be in accordance with MIL-G-10924. Failure to comply with 3~1L-G-10924 shall be cause for rejection of the mount.

4.5.2.9 Gas. Inspection and tests for nitrogen shall be in accordance with BB-N-411. Failure to comply with BB-N-411 shall be cause for rejection of the mount.

4.5.2.10 Visual examination. Each mount shall be examined for the following defects. One or more defects shall be cause for rejection of the -mount.

a. Incorrect assembly or missing parts
b. Discontinuity or absence of protective finish
c* Incorrect or missing marking
d. Color of paint not as specified
e. Workmanship not as specified.

4.5.2.11 Packaging and packing. Examination of packaging and packing shall be in accordance with Packaging Data Sheet P10891937 or P12253872 as applicable.

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4.6 Testing.

4. 6.1 Manometer. Each mount shall have a manometer test performed as descried in test method 4.7.1 after assembly, mechanical gymnasticaton, function firing and proof firing to determine compliance with nitrogen pressure requirements and the friction requirements of the floating piston, recoil rod and counter recoil piston rod. Readings shall be recorded to determine compliance with the requirements specified in 3.4.8.

4.6.2 Mechanical gymnastication. Each mount shall have a mechanical gymnastication in accordance with test method 4.7.2 to determine compliants with requirements specified in 3.4.1.

4.6.3 Functional firing. Each mount shall be fired in accordance with test method 4.7.3 to determine compliance with requirements specified in 3.4.2.

4.6.4 Proof firing. Mounts selected as specified in 4.5.1 shall be proof fired as specified in 4.7.4 to determine compliance with requirements specified in 3.4.3.

4.6.4.1 Gun mount failure. In the event a Gun Mount sample fails to satisfy acceptance criteria, the following procedure will be implemented:

a. The failed sampler accompanied with a detailed report as to the deficiencies found, will be returned to the contractor for immediate corrective action to the represented lot and subsequent production. No dismantling or altering of the mount will occur at the Government test facility without the documented consent of the contractor.

b. The corrected lot shall not be released until such time as three additional mounts selected by the Government consecutively satisfy acceptance criteria (3.4.4) when tested at the Government proving ground.

c. Three samples from the following production lot will be submitted to the Government proving ground for acceptance testing.

(1) Acceptance of these three (3) mounts shall permit return to the normal sampling procedure of one mount from each two month's production or one mount from each 25 mounts produced, whichever occurs first.

(2) Rejection of any one of the three (3) mounts tested shall requre reting of steps a, b and c.

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4. 6.5 Data requirements. The contractor shall submit a quality inspection test, Demonstration and Evaluation report in accordance with Data Item Description DI-R-1724 (tailored) giving the detailed test results for 4.6.1, 4.6.2, and 4.6.3 and 4.6.4.

4.7 Methods of inspection.

4.7.1 Manometer test.

4.7.1.1 Floating - piston and nitrogen pressure.

Procedure:

NOTE : Mechanism must be acclimated to room temperature. Prior to testing, insure proper movement of the oil index pin by draining and refilling the reserve oil. The index pin should move in when draining oil, and move out as the reserve is established. Index pin failure generally occurs on the outward movement.

(1) Insert two quick disconnects into port holes of manifold on left side of mount.

(2) Attach adapters, gage and oil line to top port manifold. Attach to oil pump. Purge lines of air prior to final tightening of connections. Tighten all connections upon completion of purging.

(3) Place handle of rotary control valve in “return” position.

(4) Slowly pump oil into the recuperator cylinder until oil index pin is fully extended (approximately 3/16” from outside face of head). At this point, while slowly applying the hand stroke to pump, the high reading should be recorded. This reading represents the nitrogen pressure (AP) plus the floating piston friction (F).

(5) Slowly open bleeder valve to allow oil to escape by drops. Note droppage of gage hand. When gage hand comes to a balance, record this reading. This reading represents the nitrogen pressure (AP) minus the floating piston friction (F).

(6) Repeat the operations (outlined in (4) and (5) above)), until three sets of nearly uniform readings are obtained. Average the three sets of readings for AP + F ((4) above) and AP-F ((5) above) and use these figures for calculations.

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(7) Initial nitrogen pressure for mount assembly is 2300 psi at 70°F with a ± psi variation per each degree of temperature variation (+4 psi per degree over 70°F; -4 psi per degree under 70°F).

(8) To determine floating piston friction, use readings obtained at (4) and (5) above and apply to the following formula (record on shop record):

$$\text{Piston Friction} \qquad \frac{(AP + F) - (AP - F)}{2}$$

Tolerances: High - 250 psi
Low - 30 psi

(9) To determine nitrogen pressure, use readings obtained at (4) and (5) above and apply to the following formula (record on Shop Record):

$$\text{Nitrogen Pressure} = \frac{(AP+F)+(AP-F)}{2}$$

Tolerance: This reading should be the same as or up to 30 psi higher than the initial direct reading.

4.7.1.2 Counter-recoil rod and stuffing box friction.

Procedure:

(1) Remove head assembly from front of counter-recoil cylinder. Screw nut of screw jack into counter-recoil cylinder. Install the remaining equipment using the same procedures outlines in 4.7.1.1.

(2) Open valve at bottom of oil gage adapter; jack rod out-of-battery 5 inches.

(3) Close bleeder valve on pressure gage adapter. Release -jack and pump rod back into the in-battery position. Exercise by repeating above operation 3 times.

(4) The fourth time, jack rod out and hold. Close bleeder valve on pressure gage adapter and increase pressure to 600 psi. Release screw jack entire length of travel. When rod returns to the in-battery position, obtain and record gage reading.

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(5) This reading represents the friction between the counter-recoil rod and stuffing box. Repeat operation 3 times and record each reading obtained. Add the 3 readings and divide by 3 to obtain the average reading (record on Shop Record).

Tolerance Limits: Low - 50 psi
High - 250 psi

4.7.2 Mechanical Gymnastication.

Procedure:

(1) Install the mount on a government-approved gymnasticator capable of applying a force in excess of 32,000 pounds while extending the mount to the short and long recoil requirements. Remove head assembly from front of counter-recoil cylinder.

(2) Assemble M113A1, M2A2, or M201 Tube (or simulated equivalent) to mechanism. Tighten mount and tube assembly securely.

(3) with assembly at O° elevation, the rods shall be retracted and allowed to return to in-battery position for four cycles.

(4) *The* first cycle shall be at 15 inches of retraction and the subsequent three cycles shall be 30, 45 and 60 inches of retraction respectively.

(5) After completion of the four initial cycles, the rods shall be held in the retracted position to allow visual inspection of rods (rear of mount) and counter-recoil cylinder (front of mount) for scoring. Any evidence of scoring shall be cause for rejection of the mount. This examination shall be repeated after the 5th, 10th, 20th, 30th, 38th and 40th cycle.

(6) The mount shall be subject to mechanical (shop) qymnastication as follows:

38 strokes at 0° elevation - 60 to 71 inches of retraction

2 strokes (39th, 40th) at 650 elevation- 25 to 32 inches of retraction.

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The gymnasticator shall be released for mount actuated return-to-battery from maximum retraction in the 10th, 20th, 30th, 38th and 40th strokes. The maximum counter-recoil time shall be 3.5 seconds at 0^0 elevation and 4.0 seconds at 650 elevation after breakin. The mount shall operate in counter-recoil without erratic movement and with no indication of oil leakage.

(7) After gymnastication, oil shall be drained from the assembly for particle count (3.3.1).

(8) Remove tube (or simulated equivalent) and mount from gymnasticator and install head assembly.

(9) Repeat manometer operation as described in paragraph 4.7.1 (record on Shop Record).

(10) After a 24 hour observation period, repeat manometer test (record on Shop Record).

4.7.3 Functional firing. Each gun mount shall be subjected to the hydraulic artillery simulator test as specified in Table II with the recuperator nitrogen pressure set to the pressure listed in Table III prior to firing. All testing shall be done on a test stand at zero elevation with the variable recoil locked for long recoil or short recoil **as** applicable to simulate specified elevations. The simulator shall be varied as required to meet the stated rod pull requirements. Impulse applied to the recoil mechanism at the 100% acceptance level shall be 18,300 lb. sec. minimum in the long recoil mode and the short recoil mode without elevation simulation, and 21,600 lb. sec. minimum in the short *recoil* mode with elevation simulation. *At least one cylce in both* long and short recoil modes must meet or exceed the 100% impulse levels. All measurements required by 3.4.2 shall be recorded for each firing. Observation shall be made for erratic motion in counter recoil. After firing, each mount shall be examined to determine compliance with oil leakage requirements specified in 3.4.5.

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TABLE II. Functional firing.

Round Number	Siimulated Elevation	Recoil Mode	Total Required Rod Pu1l (lbs]
1	0°	Long	Final Warm Up
2	(·)0	Long	126,000 Max.
3	Maximum	Short	212,000 Max.
4	Maximum	Short	212,000 Max.
5	0°	Long	126,000 Max.

TABLE III. Recoperator nitrogen pressure.

Cannon	Nitrogen Pressure
M113/M113A1	2300 ± 25
M2A2	1900 ± 25
M201	2300 ± 25

4.7.4 Proof firing. Mounts selected as specified in 4.5.1 shall be proof fired as specified in Table IV. Recuperator nitrogen pressure shall be set at 2300 ± 25 psig prior to firing. All measurements required by 3.4.3 shall be recorded for each firing. Hydraulic pressure supplied to the gun mount shall be 2750 + 100 psig. If a M110 vehicle is used, both traversing and elevating drive assembly slip clutches shall be adjusted per TM 9-2300-216-34.2. All measurements required by 3.4.3 shall be recorded for each firing. Observation shall be made for erratic motion in counter recoil. After firing, each mount shall be examined to determine compliance with oil leakage requirements specified in 3.4.5.

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TABLE IV. Proof firing.

Round Number	Simulated Elevation	Recoil Mode	Charge
1	0	Long	M2, Zone 6
2	0	Long	M188, M188A1, Zone 8
3	Maximum	Short	M188, M188A1, Zone 8
4	Maximum	Short	M188, M188A1. Zone 8
5	0	Long	M188, M188A1, Zone 8

NOTE : All M188, M188A1 charges shall be conditioned for 24 hours minimum at 145°F ± 5°F prior to firing. A projectile weighing 242 ± .5 lbs. shall be used. The M201 cannon or M201A1 cannon without muzzle brake shall be used during proof firing.

4.8 Inspection equipment. See MIL-W-63150.

5. PACKAGING

5.1 Preservation, packaging, packing and marking. The mounts shall be unit packaged, packed and marked in accordance with the requirements of the applicable Special Packaging Instruction to the level of protection specified (see 6.2).

6. NOTES

6.1 Intended use. This type of hydropneumatic gun mounts are designed for use in combat vehicles.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Title, number and date of applicable documents and drawings.
c. Requirement and the number of mounts to be submitted for first article by the contractor (see 3.1 and 4.4).
d. Place of inspection and test.
e. Levels of preservation, packaging and packing (see 5).
f. Air conditioning for assembly (see 6.3)

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6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirement List (CDRL) , the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9(n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph No,	Date Requirements	Aplicable DID
4.6.1, 4.6.2, 4.6.3, 4.6.4	Quality Inspection Test, Demonstration and Evaluation Rpt.	DI-R-1724

6.3 Assembly areas. Assembly areas, special provisions, and equipment should be provided by the contractor. The assembly areas should be air conditioned using an air conditioning unit with a 50 micron filter. The area temperature should be maintained between 68° and 72°F. The air flow rate per cubic inch per minute should be such as to maintain area air pressure at between .75 and 1.5 inches of water. Relative humidity should not exceed 40-50 percent. Areas should be clean and free of dust, lint, and abrasive materials that may become entrapped in the assembled mechanisms. No grinding, filling, scraping, or any operation producing abrasive dust should be performed in the assembly area.

6.4 Functional firing. Information regarding the artillery simulator, see 4.7.3, is available from the Commanding Officer, Rock Island Arsenal, Rock Island, IL 61299-6000.

6.5 Supersession data. This specification includes the requirements" of Rock Island Purchase Description RIAPD-680, dated 15 July 1967.

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6.6 Subject term (key word) listing.

Construction
Data item description
First article
Functional firing
Manometer test
Mechanical gymnastication
pilot pack
proof firing
protective finishes
Quality conformance inspection

Custodian:
Army - AR

Preparing activity
Army - AR

(project 1025 -A134)

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c. Reason/Rationale for Recommendation:

8. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-63995	MOUNT, HOWITZER AND GUN, M158 and M174

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

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